DEPI EGYPT – POWER BI SPECIALIZATION

E-Commerce Sales Analytics Dashboard

Project Documentation

Sections 1 – 4

1. Project Planning & Management

1.1 Project Proposal

Project Overview

The E-Commerce Sales Analytics Dashboard is a Power BI-based data analytics project developed by a team of six students enrolled in the DEPI Egypt Power BI specialization program. The project addresses the growing need for data-driven decision-making in the retail and e-commerce sector by transforming raw transactional data into actionable business intelligence.

Objectives

- Develop an interactive Power BI dashboard that enables stakeholders to monitor sales performance across multiple dimensions including time, geography, category, and customer segment.
- Identify revenue trends, seasonal patterns, and top-performing products or regions using historical order data.
- Analyze customer behavior by segment (New, Regular, Premium, VIP) to support targeted marketing and retention strategies.
- Evaluate order fulfillment and shipping efficiency to highlight operational bottlenecks.
- Present profit margin analysis by product category and sub-category to guide pricing and inventory decisions.

Project Scope

In Scope:

- Data ingestion, cleaning, and transformation of the e-commerce sales dataset (10,000 records, 26 columns).
- Power BI data model design including relationships, calculated columns, and DAX measures.
- Dashboard development with at least 4 report pages covering: Sales Overview, Product Performance, Customer Analysis, and Shipping & Operations.
- Publishing the final dashboard to Power BI Service.

Out of Scope:

- Real-time data streaming or live database connections.
- Custom web application development.
- Predictive machine learning models (beyond Power BI built-in analytics).

1.2 Project Plan

Timeline & Milestones

Phase	Deliverable	Student Deadline	Status
1. Planning & Management	Proposal, Gantt Chart, Risk Plan	20 Feb 2026	Completed
2. Literature Review	Research Review, Grading Criteria	20 Apr 2026	Completed
3. Requirements Gathering	User Stories, Functional Specs	20 Apr 2026	Completed
4. System Analysis & Design	ERD, DAX Model, Dashboard Wireframes	1 May 2026	Completed
5. Implementation	Power BI Report Files, Source Code	10 Jul 2026	Pending
6. Final Presentation	PPT, User Manual, Video Demo	17 Jul 2026	Pending

Gantt Chart Summary

A detailed Gantt chart has been prepared in Microsoft Excel and is available in the project GitHub repository under /docs/gantt_chart.xlsx. Key milestones are:

- Week 1–4: Project kick-off, data exploration, proposal finalization.
- Week 5–8: Literature review, requirements documentation.
- Week 9–12: Data modeling, DAX development, wireframe design.
- Week 13–20: Dashboard implementation, testing, and refinement.
- Week 21–22: Final presentation, video demo, and GitHub submission.

1.3 Task Assignment & Roles

Team Member	Role	Responsibilities
Mohamed Wael Hasanin	Project Manager	Overall coordination, timeline management, stakeholder communication, GitHub repository management.
Fady Nasser	Data Engineer	Data cleaning, transformation in Power Query, schema design, data quality checks.
Yousef Mohamed	Data Analyst – Sales	Sales performance analysis, DAX measures, Sales Overview and Product Performance pages.
Mohamed Wael Mohamed	Data Analyst – Customers	Customer segmentation analysis, DAX measures, Customer Analysis report page.
Abanoub Hany	Data Analyst – Operations	Shipping and fulfillment analysis, Operations report page, KPI tracking.

Team Member	Role	Responsibilities
Mahmoud	Report Designer / QA	UI/UX design, visual consistency, final report formatting, testing and quality assurance.

1.4 Risk Assessment & Mitigation Plan

Risk	Probability	Impact	Mitigation Strategy
Data quality issues (missing values, inconsistent formats)	Medium	High	Apply Power Query transformations; document all cleaning steps; validate outputs against source.
Team member unavailability or workload conflicts	Medium	Medium	Distribute tasks clearly; maintain updated GitHub repo; weekly sync meetings.
DAX formula complexity / performance issues	Medium	High	Break complex measures into intermediate steps; use variables; test with sample data first.
Scope creep (adding extra visuals or features)	High	Medium	Strictly follow the requirements document; change requests require team approval.
Power BI Service publishing issues	Low	High	Test publishing early; keep .pbix backup file; use free Power BI account as fallback.
Dataset not representative / too small	Low	Medium	Dataset has 10,000 rows across 4 regions and 5 categories — sufficient for meaningful insights.

1.5 KPIs (Key Performance Indicators)

The following KPIs will be used to measure the success of both the project process and the dashboard output:

Project Process KPIs

- **On-Time Delivery Rate:** Percentage of phases delivered on or before the student deadline. Target: 100%.
- **GitHub Commit Frequency:** Minimum 2 commits per week per active contributor. Target: Consistent history throughout the project.
- **Documentation Completeness:** All 7 required documentation sections submitted per guidelines. Target: 100%.

Dashboard Quality KPIs

- **Report Load Time:** Time for the Power BI report to fully render. Target: < 5 seconds.

- **DAX Measure Accuracy:** All calculated measures validated against raw data cross-checks. Target: 0 errors.
- **Visual Coverage:** All defined user stories have at least one corresponding visual. Target: 100%.
- **User Satisfaction Score:** Feedback from lecturer/evaluator. Target: Score of 4/5 or above.
- **Data Freshness:** Dashboard connected to the latest version of the dataset. Target: Dataset version v1.0 used throughout.

2. Literature Review

This section summarizes the relevant body of knowledge underpinning the E-Commerce Sales Analytics Dashboard project, covering business intelligence concepts, Power BI capabilities, e-commerce analytics practices, and evaluation criteria.

2.1 Business Intelligence & Data Analytics in E-Commerce

Business Intelligence (BI) refers to the technologies, processes, and practices that transform raw data into meaningful information for informed business decisions. In e-commerce, BI is critical for understanding customer purchasing behavior, optimizing supply chains, and improving profitability. Studies consistently show that data-driven organizations outperform peers in revenue growth and operational efficiency.

Key themes from the literature include:

- Sales trend analysis using time-series visualization to detect seasonality, growth patterns, and anomalies.
- Customer lifetime value (CLV) segmentation models that differentiate New, Regular, Premium, and VIP customers.
- Product category performance benchmarking to guide inventory and pricing strategy.
- Shipping and logistics KPIs such as average delivery time and return rates as proxies for customer satisfaction.

2.2 Microsoft Power BI as a BI Platform

Power BI is Microsoft's cloud-based business analytics service that provides interactive visualizations and self-service BI capabilities. It is widely adopted in industry for the following reasons:

- Power Query enables robust ETL (Extract, Transform, Load) operations directly within the tool.
- DAX (Data Analysis Expressions) provides a powerful formula language for calculated columns, measures, and time-intelligence functions.
- The star schema data model is the recommended approach for Power BI, separating fact tables from dimension tables for performance and clarity.
- Power BI Service allows publishing, sharing, and scheduling of data refreshes, making it suitable for enterprise deployment.

For this project, Power BI Desktop is the primary development tool, with Power BI Service used for final publishing and sharing.

2.3 Dataset Overview

The dataset used is a structured e-commerce sales dataset containing 10,000 transaction records across 26 fields. Below is a summary of the dataset dimensions:

Dimension	Values / Range
Time Period	2021 – 2023 (Multi-year, with Month, Quarter, Season columns)
Regions	North America, Europe, Asia, Middle East
Product Categories	Electronics, Clothing, Books & Media, Home & Kitchen, Beauty & Health
Customer Segments	New, Regular, Premium, VIP
Order Statuses	Delivered, Processing, Returned, Cancelled
Key Metrics	Revenue, Cost, Profit, Profit Margin %, Shipping Cost, Unit Price, Quantity, Discount
Shipping Methods	Economy, Standard, Express
Payment Methods	Credit Card, Google Pay, and others

2.4 Suggested Improvements

Based on review of similar BI projects and best practices, the following enhancements are recommended for future iterations of this project:

- Integrate a live data source (e.g., SQL Server or SharePoint) for real-time refreshes.
- Add predictive analytics using Power BI's built-in AI visuals (e.g., Key Influencers, Decomposition Tree).
- Incorporate a customer churn prediction model using Python or R scripts embedded in Power BI.
- Expand geographic analysis with map visuals and drill-through by country/city.
- Apply Row-Level Security (RLS) in Power BI Service for role-based data access control.

2.5 Final Grading Criteria

According to the project documentation guidelines, the final assessment will be based on the following breakdown:

Evaluation Area	Description	Weight
Documentation	Completeness and quality of all 7 documentation sections	25%
Implementation	Power BI report quality, DAX accuracy, data model design	35%
Testing & QA	Test plan coverage, bug identification and resolution	15%
Presentation	Clarity of final presentation, demo video, and user manual	25%

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Key Needs
DEPI Lecturer / Evaluator	Primary evaluator and project sponsor	Well-documented, functional dashboard; adherence to academic standards; clear presentation.
Team Members (×6)	Developers and analysts	Clear task assignments, shared GitHub repository, well-defined data model.
E-Commerce Business Owner (Persona)	Simulated end user of the dashboard	Quick access to revenue KPIs, product performance, and customer trends; intuitive navigation.
Marketing Manager (Persona)	Simulated end user	Customer segmentation insights, campaign performance indicators, region-level sales data.
Operations Manager (Persona)	Simulated end user	Shipping efficiency metrics, order status distribution, return and cancellation rates.

3.2 User Stories & Use Cases

User Stories

ID	As a...	I want to...	So that...
US-01	Business Owner	See total revenue, profit, and orders for any selected period	I can track overall business health at a glance.
US-02	Business Owner	Compare sales performance across quarters and years	I can identify growth trends and seasonality.
US-03	Marketing Manager	View sales breakdown by customer segment (New/Regular/Premium/VIP)	I can tailor promotions to high-value customers.
US-04	Marketing Manager	Filter data by region and country	I can assess regional market performance.
US-05	Operations Manager	Monitor average shipping days by shipping method	I can identify fulfillment delays.
US-06	Operations Manager	Track order status distribution (Delivered/Returned/Cancelled)	I can measure operational efficiency.
US-07	Business Owner	Drill down from Category to Sub-Category to Product	I can find specific underperforming products.

ID	As a...	I want to...	So that...
US-08	Marketing Manager	See discount impact on profit margin	I can optimize discount strategy.

3.3 Functional Requirements

Data & Model Requirements

- **FR-01:** The system shall ingest and clean the e-commerce CSV dataset using Power Query, handling null values, data type inconsistencies, and duplicate records.
- **FR-02:** The data model shall follow a star schema with a central fact table (Orders) and dimension tables for Date, Product, Customer, Geography, and Shipping.
- **FR-03:** The system shall calculate the following DAX measures: Total Revenue, Total Cost, Total Profit, Profit Margin %, Total Orders, Average Order Value, Average Discount Rate, and Average Shipping Days.
- **FR-04:** The system shall support time-intelligence calculations including Month-over-Month (MoM) and Year-over-Year (YoY) revenue comparisons.

Dashboard & Visualization Requirements

- **FR-05:** The dashboard shall contain a minimum of 4 report pages: (1) Sales Overview, (2) Product Performance, (3) Customer Analysis, (4) Shipping & Operations.
- **FR-06:** All pages shall include interactive slicers for Year, Quarter, Region, and Category.
- **FR-07:** The Sales Overview page shall display KPI cards for Total Revenue, Total Profit, Total Orders, and Profit Margin %.
- **FR-08:** The Product Performance page shall include a bar chart of revenue by category, a matrix of sub-category profitability, and a top-10 products table.
- **FR-09:** The Customer Analysis page shall display a donut chart of orders by segment, a bar chart of revenue by gender, and a scatter plot of revenue vs. profit by segment.
- **FR-10:** The Shipping & Operations page shall show order status distribution, average shipping days by method, and shipping cost analysis.
- **FR-11:** The dashboard shall support drill-through from Category to individual product details.

3.4 Non-Functional Requirements

- **NFR-01 – Performance:** The Power BI report shall render fully within 5 seconds on a standard laptop with the 10,000-row dataset.
- **NFR-02 – Usability:** All visuals shall include clear titles, axis labels, and tooltips. Color scheme shall meet WCAG AA contrast guidelines.
- **NFR-03 – Reliability:** All DAX measures shall produce consistent results validated against raw CSV calculations with zero discrepancies.

- **NFR-04 – Maintainability:** Power Query steps shall be named descriptively. DAX measures shall be organized in a dedicated Measures table with clear naming conventions (e.g., [Total Revenue], [YoY Revenue %]).
- **NFR-05 – Portability:** The .pbix file shall be compatible with Power BI Desktop version 2.120 or later and publishable to Power BI Service Free tier.
- **NFR-06 – Security:** The published report shall be accessible via a shareable link; no personally identifiable information (PII) is present in the dataset.

4. System Analysis & Design

4.1 Problem Statement & Objectives

Problem Statement

E-commerce businesses generate large volumes of transactional data across products, customers, regions, and time periods. Without a structured analytics solution, decision-makers cannot efficiently identify sales trends, understand customer behavior, or detect operational inefficiencies. Spreadsheet-based analysis is time-consuming and lacks interactivity, making it unsuitable for ongoing business monitoring.

Project Objectives

- Build a self-service Power BI dashboard that consolidates all e-commerce transaction data into a single, interactive reporting solution.
- Enable drill-down analysis from high-level KPIs to granular product and customer details.
- Provide time-intelligence comparisons (monthly, quarterly, yearly) for trend identification.
- Support multi-dimensional filtering so stakeholders can slice data by region, category, segment, and time without technical knowledge.

4.2 Database Design & Data Modeling

Star Schema Design

The Power BI data model follows a star schema architecture, which is the industry best practice for analytical workloads. The schema consists of one fact table and five dimension tables:

Table	Type	Key Columns	Description
fact_Orders	Fact	Order_ID (PK), Date_Key, Product_Key, Customer_Key, Geo_Key, Shipping_Key	Central transaction table with all numeric measures: Revenue, Cost, Profit, Quantity, Discount, Shipping Cost.
dim_Date	Dimension	Date_Key (PK), Order_Date, Year, Month, Quarter, Season	Time dimension enabling all time-intelligence DAX functions.
dim_Product	Dimension	Product_Key (PK), Product_Name, Category, Sub_Category, Unit_Price	Product hierarchy enabling Category > Sub-Category > Product drill-down.
dim_Customer	Dimension	Customer_Key (PK), Customer_ID, Customer_Gender, Customer_Segment	Customer attributes for segmentation analysis.

Table	Type	Key Columns	Description
dim_Geography	Dimension	Geo_Key (PK), Region, Country	Geographic hierarchy for regional and country-level filtering.
dim_Shipping	Dimension	Shipping_Key (PK), Shipping_Method, Payment_Method, Order_Status	Operational dimension for shipping and fulfillment analysis.

Entity-Relationship Overview

All dimension tables connect to the fact_Orders table via one-to-many relationships (1:*), with the single side on the dimension tables. Cross-filter direction is set to Single (dimension to fact) to prevent ambiguity and ensure optimal query performance. The dim_Date table is marked as a Date Table in Power BI to enable time-intelligence functions.

4.3 DAX Measures Design

Core Measures

Measure Name	DAX Formula (Simplified)	Purpose
Total Revenue	SUM(fact_Orders[Revenue])	Aggregate revenue across all orders.
Total Cost	SUM(fact_Orders[Cost])	Aggregate cost across all orders.
Total Profit	[Total Revenue] - [Total Cost]	Net profit calculation.
Profit Margin %	DIVIDE([Total Profit], [Total Revenue], 0)	Profitability ratio.
Total Orders	COUNTROWS(fact_Orders)	Count of all transactions.
Avg Order Value	DIVIDE([Total Revenue], [Total Orders], 0)	Average revenue per order.
Avg Discount Rate	AVERAGE(fact_Orders[Discount])	Mean discount applied.
Avg Shipping Days	AVERAGE(fact_Orders[Shipping_Days])	Average fulfillment time.
YoY Revenue %	DIVIDE([Total Revenue] - [Prev Year Revenue], [Prev Year Revenue], 0)	Year-over-year growth rate.
MoM Revenue %	DIVIDE([Total Revenue] - [Prev Month Revenue], [Prev Month Revenue], 0)	Month-over-month growth rate.

4.4 Data Flow & System Behavior

Data Flow Description

The data pipeline for this project follows a linear ETL flow within Power BI:

Step	Stage	Tool	Description
1	Extract	Power BI Desktop	Import ecommerce_sales_dataset.csv as the raw data source.
2	Transform	Power Query Editor	Remove nulls, correct data types, rename columns, create surrogate keys, build dimension tables by referencing and deduplicating the source query.
3	Load	Power BI Data Model	Load cleaned fact and dimension tables into the in-memory VertiPaq engine.
4	Model	DAX / Model View	Define relationships, create calculated columns (e.g., Date_Key as YYYYMMDD integer), create all DAX measures in a dedicated Measures table.
5	Visualize	Power BI Report View	Build 4 report pages with interactive visuals consuming the DAX measures.
6	Publish	Power BI Service	Publish the .pbix file to Power BI Service for sharing and access via shareable link.

4.5 Dashboard UI/UX Design

Report Pages & Visual Layout

Page	Key Visuals	Purpose
1 – Sales Overview	4× KPI Cards (Revenue, Profit, Orders, Margin), Line Chart (Revenue over Time), Bar Chart (Revenue by Region), Slicers (Year, Quarter, Region)	High-level executive summary of business performance.
2 – Product Performance	Bar Chart (Revenue by Category), Matrix (Sub-Category × Profit Margin), Top-10 Products Table, Treemap (Revenue share), Slicer (Category)	Identify best and worst performing products.
3 – Customer Analysis	Donut Chart (Orders by Segment), Bar Chart (Revenue by Gender), Column Chart (Avg Order Value by Segment), Scatter Plot (Revenue vs Profit), Slicer (Segment)	Understand customer base and segment value.
4 – Shipping & Operations	Stacked Bar (Order Status), Clustered Bar (Avg Shipping Days by Method), Card (Return Rate %), Line Chart (Shipping Cost trend), Slicer (Shipping Method)	Monitor operational efficiency and fulfillment quality.

UI/UX Design Guidelines

- **Color Scheme:** Primary blue (#1F4E79 and #2E75B6) for headers and accents; light gray (#F2F2F2) for page background; white (#FFFFFF) for visual backgrounds. Consistent with Microsoft Fluent Design.
- **Typography:** Segoe UI (Power BI default); Title: 16pt Bold; Visual Title: 12pt Bold; Labels: 10pt Regular.
- **Layout:** All pages follow a consistent grid: KPI cards in the top row, main chart in the center, supporting visuals on the right/bottom, slicers on the left panel.
- **Accessibility:** All visuals include alt-text descriptions; color coding is supplemented with labels to accommodate color-blind users.
- **Navigation:** Page navigator buttons are placed in a consistent left-side panel for easy multi-page navigation.

4.6 Technology Stack

Layer	Technology	Purpose
Data Storage	CSV File (ecommerce_sales_dataset.csv)	Source data file, hosted in GitHub repository.
ETL / Data Prep	Power Query (M Language)	Data cleaning, transformation, and dimension table creation.
Data Model	Power BI Desktop – VertiPaq Engine	In-memory columnar store; star schema relationships.
Analytics Layer	DAX (Data Analysis Expressions)	KPI measures, time-intelligence, calculated columns.
Visualization	Power BI Desktop Report View	Interactive charts, tables, KPI cards, slicers, drill-throughs.
Publishing	Power BI Service (Free Tier)	Report sharing via web link; workspace management.
Version Control	GitHub	All .pbix files, documentation, and dataset versioning.
Documentation	Microsoft Word / PDF	Project documentation as per DEPI guidelines.